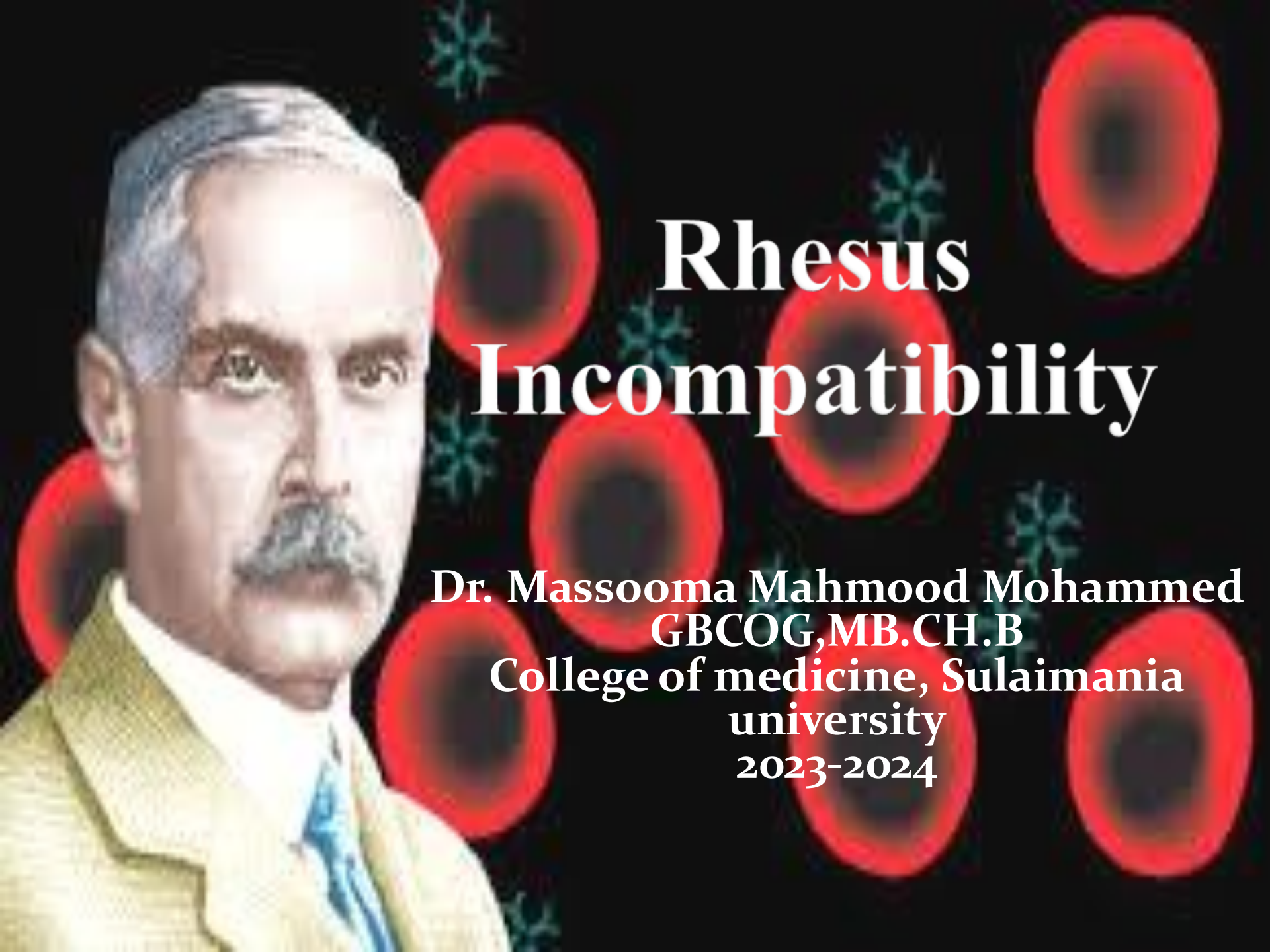




Rhesus Incompatibility

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Introduction:

Blood group is defined in two ways:

- 1-ABO group, allowing four different permutations of blood group (**O, A, B , AB**).
- 2-Rhesus system, which consists of **C, D and E antigens.**
- 3.Also non Rhesus groups such as Kell , Duffy (Fy)and Kidd (Jk) exist

Introduction

The importance of these blood group systems occur when there is a mismatch between the fetus and mother when fetal red cells pass across to maternal circulation, as they do during pregnancy .

*Sensitization of the maternal immune system to these fetal (**foreign**) red blood cells may occur and subsequently give rise to the **haemolytic disease of the fetus and newborn (HDFN)**.

▪

The Inheritance of Rhesus system:

The Rhesus system is coded on two adjacent genes, which sit within chromosome **one** .

One gene codes for antigen polypeptides C/c and E/e while the other codes for the D polypeptide (Rhesus antigen).

Notes that the d (little d) antigen has not been identified so it may be that women who are D negative lack the antigen altogether.

The Inheritance of Rhesus system:

Antigen expression is usually dominant ,
whereas those who have a negative
phenotype are either homozygous for the
recessive allele or have a deletion of that
gene.

In practice, only **anti-D and anti-c** regularly
cause HDFN (haemolytic disease of fetus and
newborn and **anti-D is much more common
than anti-c.**

The aetiology of rhesus disease:

Occurrence of HDFN as a result of Rhesus isoimmunization involves three key stages :

- 1-First**, a Rhesus negative mother must conceive a baby who has inherited the rhesus positive phenotype from the father .
- 2-second**, fetal cell must gain access to the maternal circulation in a sufficient volume to provoke a maternal antibody response .
- 3-finally**, maternal antibody must gain transplacental access and cause immune destruction of red cells in the fetus.

Why Does Rh Status Matter?

Fetal RBC cross to maternal circulation

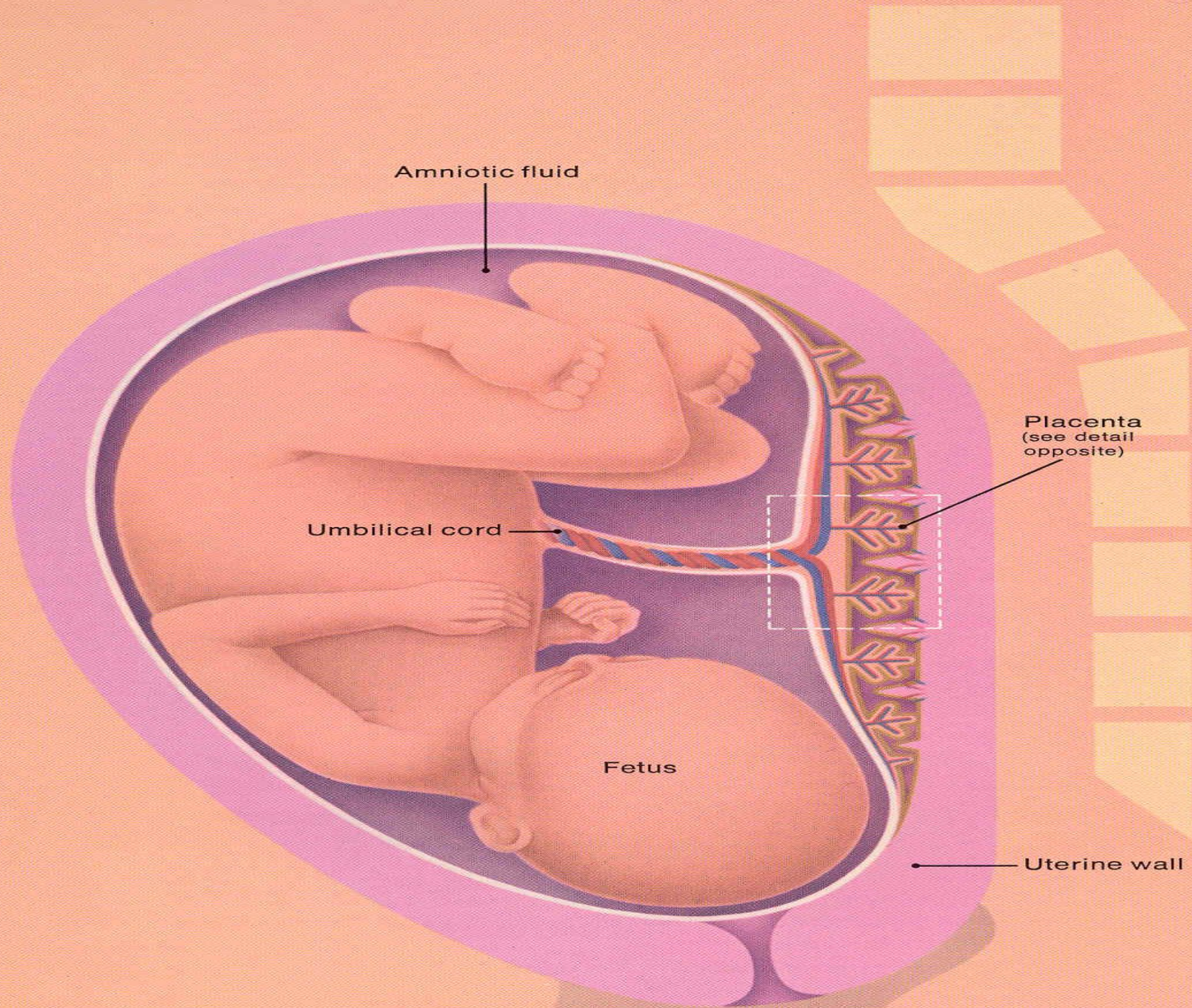
Maternal immune system recognizes foreign antigens if fetus Rh + and mother Rh –

Antibodies are formed against fetal antigens

**Subsequent pregnancy with Rh+ fetus,
immune system activated
and large amounts of Ab formed**

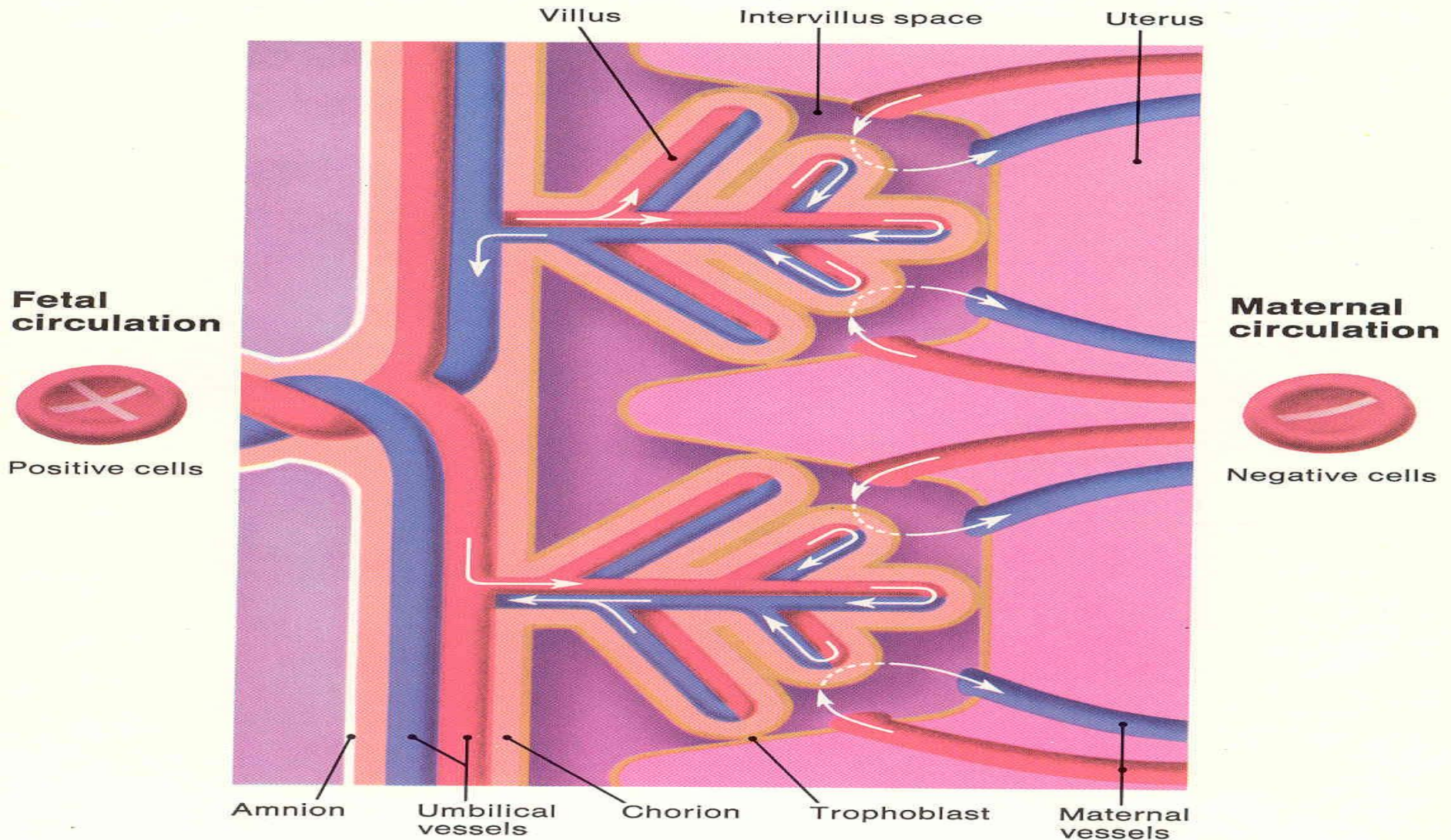
IgG Ab cross placenta & attack fetal RBC

Fetal anemia, hydrops, etc



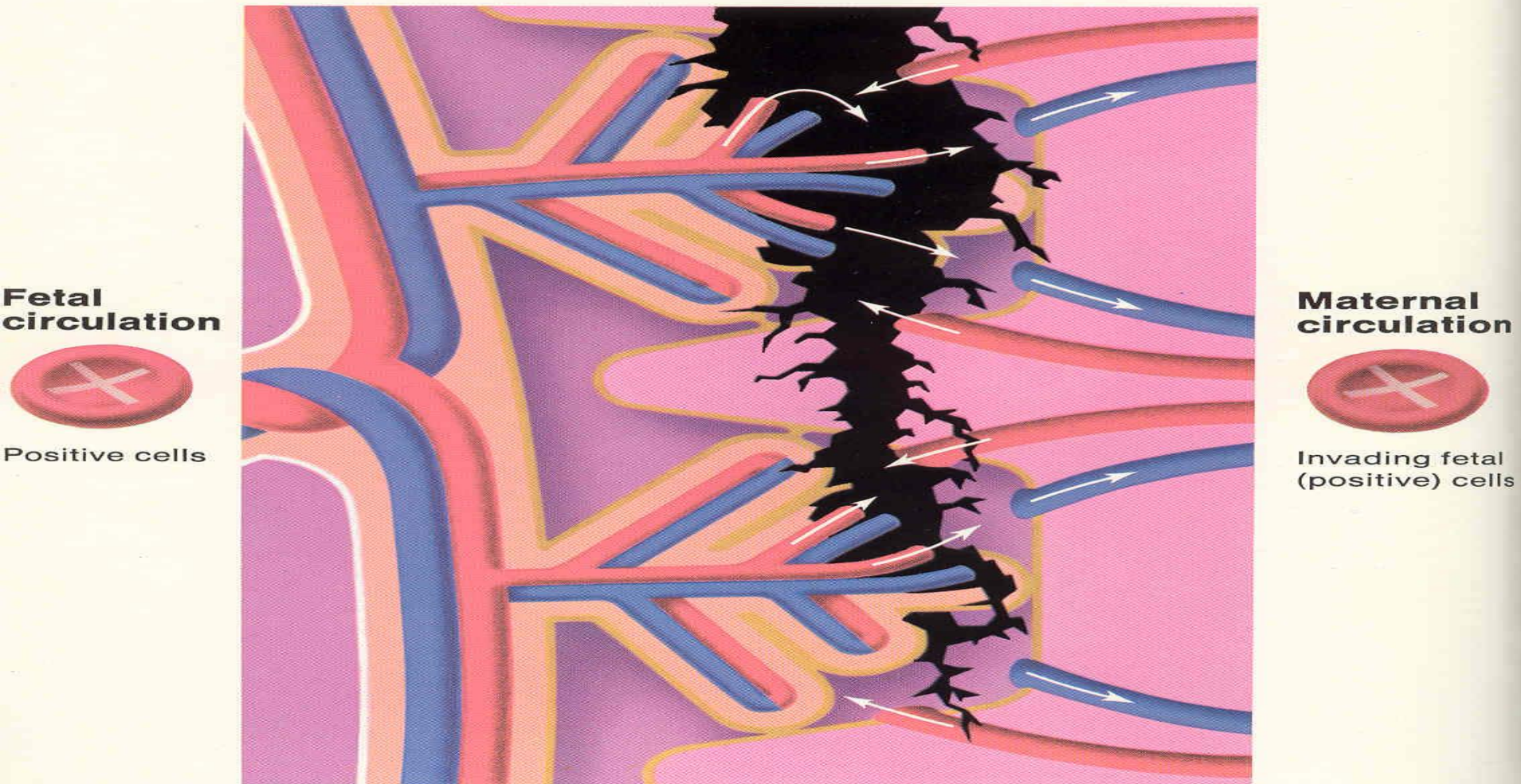
Scheme of Placental Circulations

White arrows depict separate routes of fetal and maternal circulations within the placenta. Dotted lines represent oxygen, nutrient and waste exchange through the placental barrier.

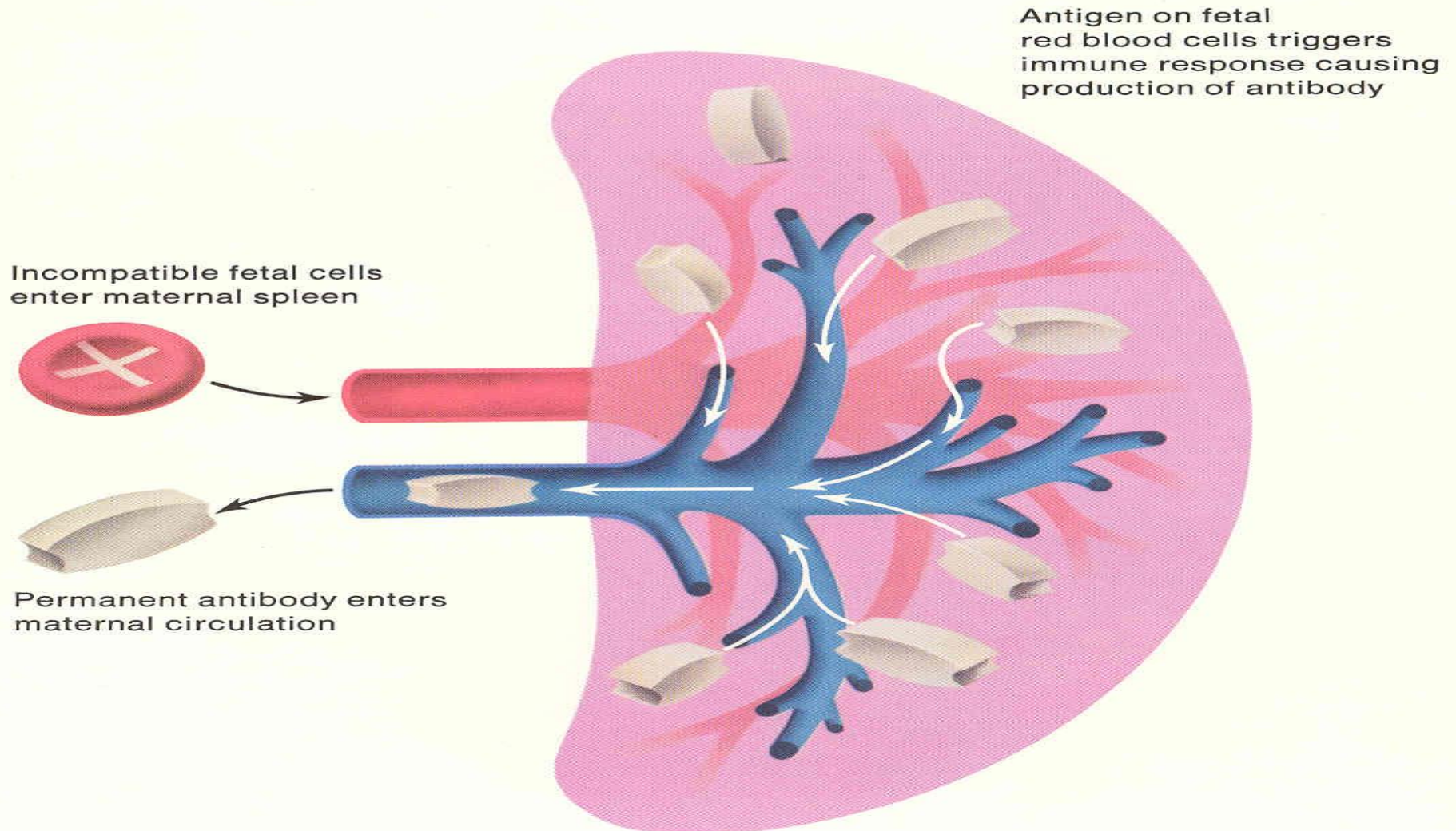


Separation of Placenta Following Delivery

Diagram portrays the rupture of placental vessels (villi) and connective tissue allowing escape of fetal blood cells. Prior to complete constriction of open-end maternal vessels, some fetal blood may enter maternal circulation.



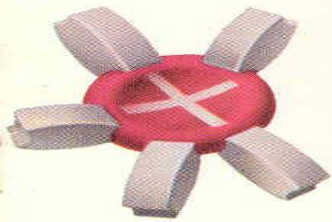
Mother's Spleen After Delivery of Incompatible Infant



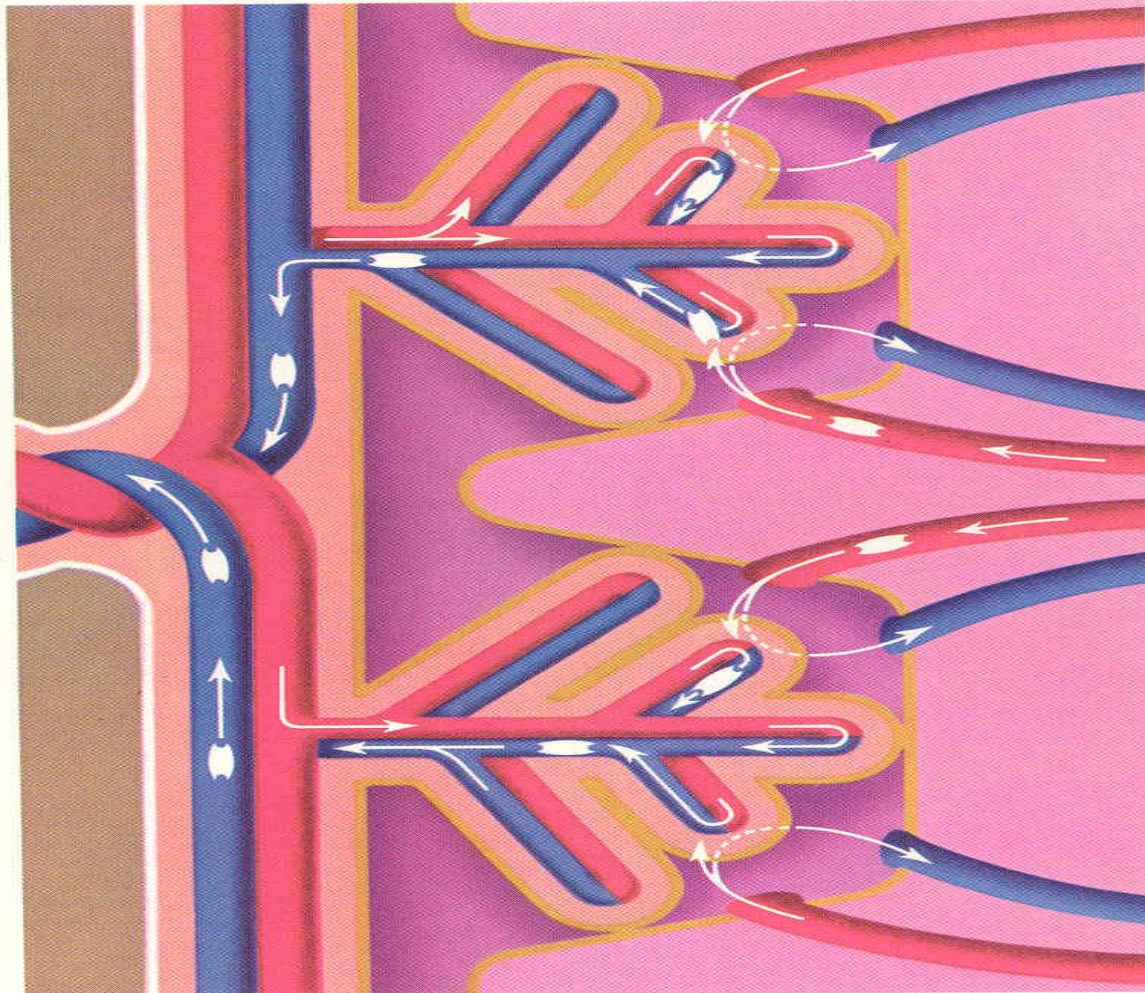
Subsequent Incompatible Pregnancy

Residual antibodies produced as a response to red cells of a previous incompatible fetus or donor are transported through the placental barrier. They attach to the specific red cell antigen sites of the incompatible fetus of the current pregnancy. Sensitized cells do not have a normal life span; the baby suffers from anemia and its consequences.

Fetal circulation



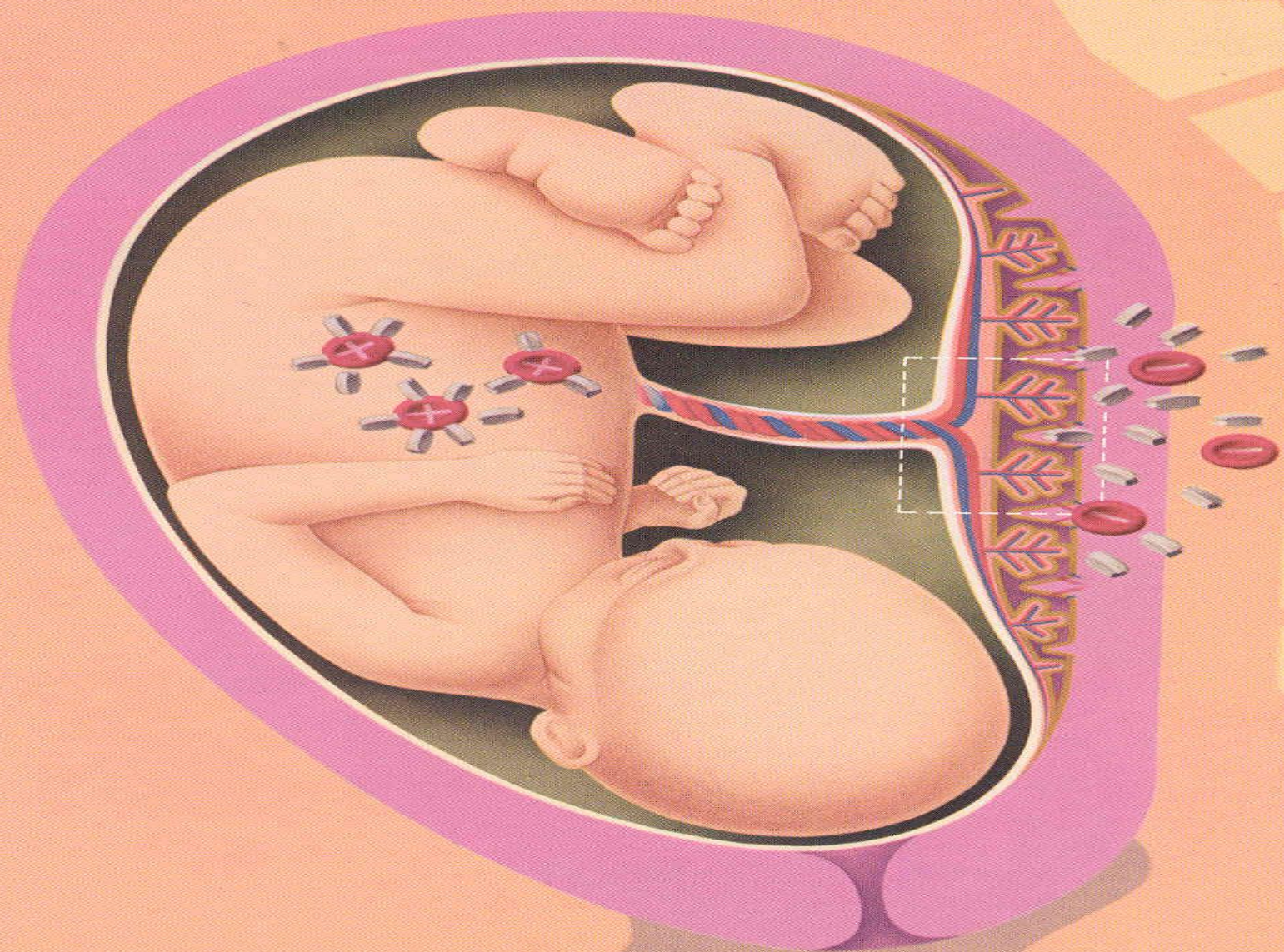
Maternal antibodies attach to incompatible fetal cells



Maternal circulation



Circulating maternal antibodies produced by previous isoimmunization enter placenta



Rhesus disease

-Rhesus disease does not affect a first pregnancy.

as the primary response is weak and consists of **IgM** antibodies that do not cross the placenta .

there after **IgG** antibodies are produced and cross the placenta .

-Rhesus antigen are well expressed by the fetus as early as 30 days gestation . so in a subsequent pregnancy, when maternal resensitization occurs (**rhesus positive red cells pass from the baby to the maternal circulation**) **IgG** antibodies cross from the mother to the fetal circulation.

-If these antibodies are present in sufficient quantities **fetal haemolysis** may occur ,leading to severe anaemia that the fetus may die unless transfusion is performed.

Causes of Fetomaternal Hemorrhage Associated with Red Cell Antigen Alloimmunization

Pregnancy loss : ectopic pregnancy
spontaneous abortion
elective abortion
fetal death (any trimester)

Procedure : chorionic villous sampling
amniocentesis
fetal blood sampling
evacuation of molar pregnancy

Delivery : abdominal trauma
placental abruption
unexplained vaginal bleeding during pregnancy
manual placental removal
external cephalic version

Prevalence of Rhesus disease:

- 1-The prevalence of D Rhesus negativity is **15%** in the Caucasian population
- 2-**It is very uncommon in orientals.**
- 3-Rhesus disease is most common in countries where anti-D prophylaxis is not widespread, **as in middle east and Russia**

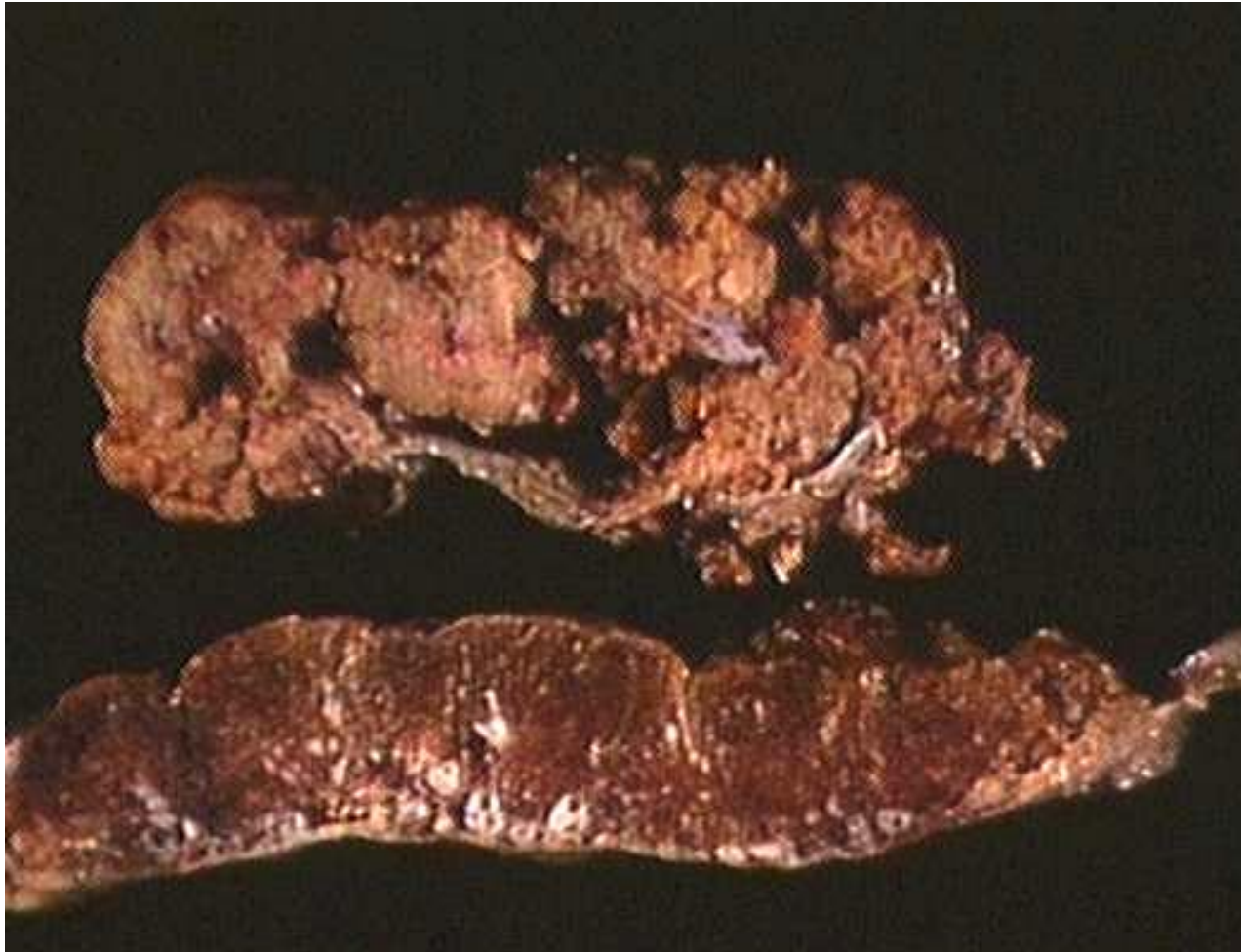
fetal sign of anaemia

- 1- Polyhydramnios.
- 2- Enlarged fetal heart.
- 3- Ascites and pericardial effusion.
- 4- Reduced fetal movements.
- 5- Abnormal CTG with reduced variability, eventually a sinusoidal trace.**
- 6- Hyperdynamic fetal circulation can be detected by ultrasound by measuring **increased velocities in the middle cerebral artery or aorta.**

Hydropic fetus



Placenta of hydropic fetus



Body wall oedema of hydropic fetus





A hydropic neonate under extensive intensive care

Management of Rhesus disease in a sensitized women:

- ❖ Once a women who is D Rhesus negative has been sensitized to the D Rhesus antigen , **no amount of anti-D will ever turn the clock back** .
- ❖ In a subsequent pregnancy , close surveillance is required .
- ❖ Rhesus disease gets **worse** with successive pregnancies.
- ❖ so it is important to note the **severity** of the disease in previous pregnancies .
- ❖ Standard management involves monitoring antibody levels **every 2 – 4 weeks** from booking.
- ❖ Antibody levels or quantity can be described using the titer or by using **IU (international units)** as a standard quantification method.

The titer of anti-D level:

Anti-D level

1. < 4 IU/mL
2. 4 – 15 IU/mL
3. > 15 IU/mL
fetalis

Outcome

HDFN unlikely
Moderate risk of HDFN
High risk of hydrops
fetalis

Clinical management for positive Antibody screen

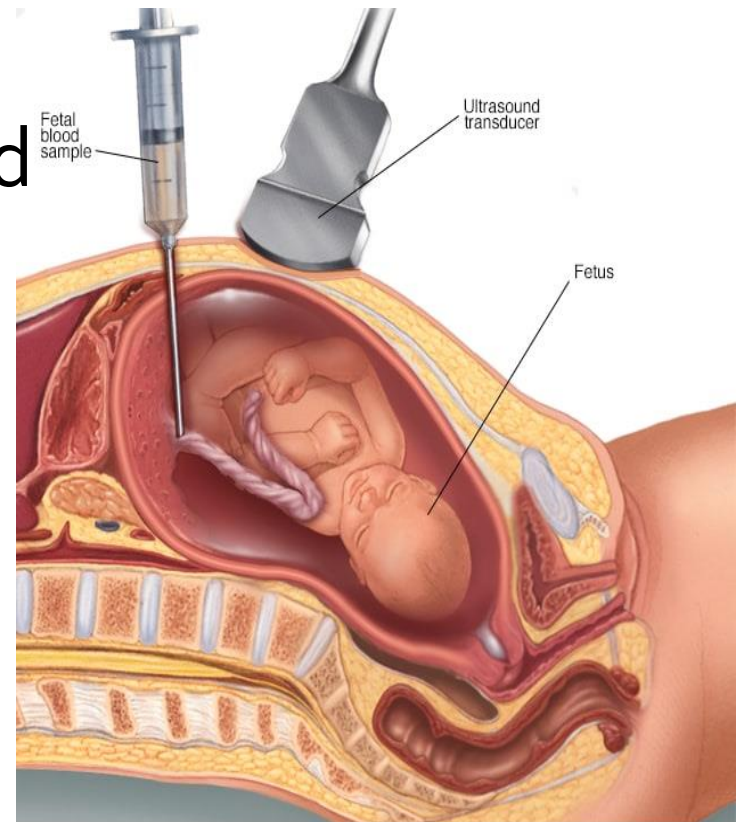
- ❖ Determine risk factors for isoimmunization:
 - Past pregnancies
 - Transfusions
 - Shared needles
- ❖ Determine father's RBC antigen status and zygosity
- ❖ If paternity unknown or father is (+) for antigen, fetus is at risk
- ❖ Consider invasive testing at titer of 1:32 or greater by indirect coombs(1:16 most often used)

Invasive methods: Amniocentesis

- Critical titre/previous affected infant
- Avoid transplacental needle passage
- Detection of fetal bilirubin through PCR test
- Bilirubin correlates with fetal hemolysis
- in the past, the **bilirubin concentration of amniotic fluid was determined optically** to give an indirect measures of fetal haemolysis by using **Liley** curve.

Invasive method: cordocentesis

Direct fetal blood sampling is the '**gold standard**' for the **diagnosis of fetal anaemia**, but carries a procedure-related risk of miscarriage. It is done in 20th -24th GW.



Non invasive methods:

1-middle cerebral artery (MCA)

Doppler's (peak velocity measurement) have been correlate reliably with fetal anaemia, **the sensitivity is reported at 100% .**

- A fetus with raised MCA dopplers has a high probability of anaemia. These cases are not common and the treatment should be in **tertiary fetal medicine centres.**

2- NIIP: cff- DNA

Management: Treatment

Treatment options include delivery or fetal blood transfusion

-delivery of the fetus is an option if the fetus is sufficiently mature, however delivery of an anaemic, rapidly haemolysing premature baby is a significant risk and should not be undertaken lightly.

Delivery must take place in:

a unit where adequate neonatal support and expertise is available and generally delivery should not be before **36-37 weeks** of gestation unless there are difficulty with fetal transfusion.

Management: Treatment

- * Once a decision has been made that the fetus is severely anaemic and requires a blood transfusion **Fetal blood transfusion is life saving in a severely anaemic fetus that is too premature for delivery to be contemplated by a Hkt<30**
- * **The aim of blood transfusion :**
 1. Restoring haemoglobin levels.
 2. Reversing or preventing hydrops or fetal death.
- * **A side effect** of blood transfusion it will suppress fetal erythropoiesis , which reduces the concentration of antigen positive cells available for haemolysis.

routes of administration:

1. Into the umbilical vein at the point of the cord insertion .
2. into the intrahepatic vein .
3. into the peritoneal cavity (**may be the only way in low gestation**).
4. into the fetal heart.

The transfused blood should be

- 1. O-RhD negative.**
- 2. Crossmatched with a maternal sample.**
- 3. densely packed (Hb around 30 g/L),so that small volumes are used.**
- 4. White cell depleted and irradiated.**
- 5. screened for infection including CMV.**

At delivery:

- if the baby is known to be anaemic or has had multiple transfusions, a **neonatologist must be present**.
- At delivery should exchange transfusion be required .
- blood must therefore always be ready for the delivery
- all babies born to Rhesus –negative women should have cord blood taken at delivery for a blood count, blood group and direct coombs test.**

Prevention

- Give 300 mcg within 72 hrs of delivery to unsensitized Rh(-) women(Rh positive infant)
- Anti-D immunoglobulins mop up any circulating rhesus-positive cells before an immune response is excited in the mother.
- **In many countries , Rhesus-negative women are given anti-D at 28 and/or 34 weeks routinely**
- **Give RhoGam for partial molar pregnancy, ectopic pregnancy, ECV, invasive procedures,,,etc.**
- The exact **dose** is determined by **gestation at which sensitization occurred and size of fetomaternal haemorrhage.**
- **kleihauer test** is done , when there is a suspicion of large amount of blood transfusion,e.g. Placenta previa, abruprtion, haemorrhage after a blunt trauma, cordocentesis,.....

Kleihauer-Betke Test

- **% fetal RBC in maternal circulation**
- Fetal erythrocytes contain **Hb F** which is more resistant to acid elution than **Hb A** so after exposure to acid, only fetal cells remain & can be identified with stain
- It will allow calculation of the amount of extra anti-D immunoglobulin required should a large transfusion have occurred.

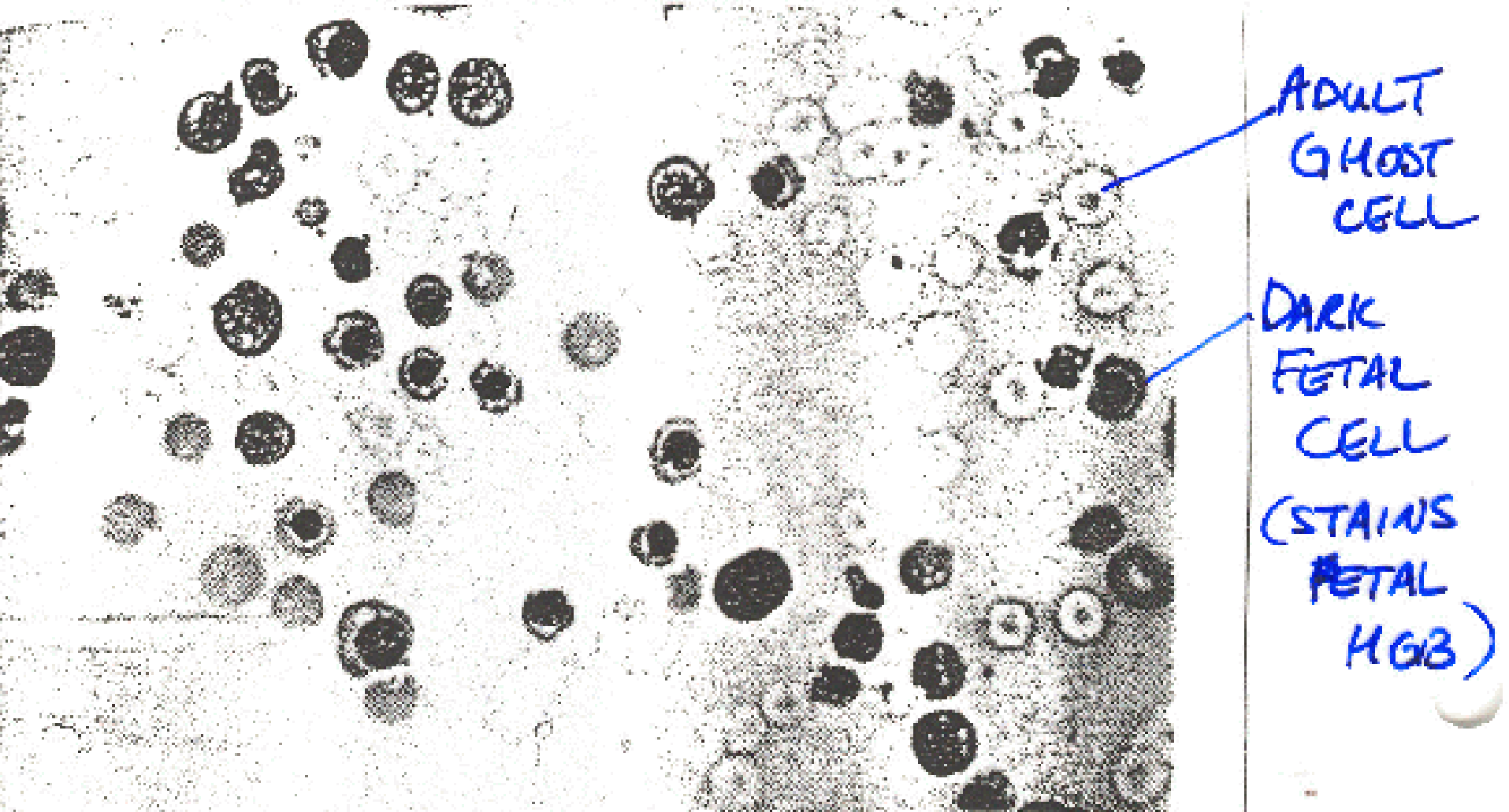


Figure 4. Cord blood of baby with severe Rh erythroblastosis fetalis who required multiple fetal transfusions and exchange transfusions. Smear treated by Kleihauer technique and Wright's stain. Note adult donor ghost red cells, dark fetal red cells, and early fetal erythroid series from erythroblasts through to normoblasts.

Thank you

